Week 48

v1.0.1

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```
try:
    import fysisk_biokemi
    print("Already installed")
except ImportError:
    %pip install -q "fysisk_biokemi[pyodide] @ https://au-mbg.github.io/fysisk-biokemi/
wheels/fysisk_biokemi-0.2.2-py3-none-any.whl"
```

---

1 Enzyme inhibitors (I)

```
import matplotlib.pyplot as plt
import pandas as pd
from scipy.optimize import curve_fit
import numpy as np
from IPython.display import display
pd.set_option('display.max_rows', 6)
```

An enzyme obeying the Michaelis-Menten kinetics model was tested for substrate conversion in the absence and presence of an inhibitor, called inhibitor1 at a concentration of $[\mathrm{I}] = 2.5 \cdot 10^{-3}$ M. The data set is contained in the file `enzyme-inhib-i.xlsx`. Using this data a researcher wanted to determine the type of inhibition.

Start by loading the dataset

```
data_file_name = ... # Write the  of the uploaded data file
df = pd.read_excel(data_file_name)
display(df)
```

(a) Convert units

Convert the concentrations of substrate and the initial velocities to units given in M and $\mathrm{M} \cdot \mathrm{s}^{-1}$, respectively.

```
## Your task: Make new columns with the properties in SI units.
df['[S]_(M)'] = ...
df['V0_no_inhib_(M/s)'] = ...
... # This one you'll have to do all on your own.
display(df)
```

(b) Plot

Plot the initial velocities of both experiments as a function of substrate concentration in one plot. Estimate K_M and $V_{\max}$ in the presence and absence of inhibitor from the plot.

```
fig, ax = plt.subplots(figsize=(6, 4))

## Your task: Plot the initial velocities for both experiments.
... # Your code for plotting here
... # and here.

## Sets the x and y-axis labels and shows the legend.
ax.set_xlabel('[S] (M)')
ax.set_ylabel('$V_0$ (M/s)')
ax.legend()
```

Use the plots to estimate K_M and $V_{\max}$ and assign in the cell below

```
## Your task: Estimate parameters without inhibitor.
K_M_no_inhib_estimate = ...
Vmax_no_inhib_estimate = ...
```

```
## Your task: Estimate parameters with inhibitor.
K_M_inhib_estimate = ...
Vmax_inhib_estimate = ...
```

(c) Fit

The researcher wanted to determine $V_{\max}$ and K_M values for both experiments in order to correctly conclude on the type of inhibitor.

Determine $V_{\max}$ and K_M by fitting.

Start writing the function to fit with

```
## Your task: Implement the Michealis-Menten equation as a function
def michaelis_menten(S, Vmax, Km):
    result = ...
    return result
```

And use the `curve_fit`-function to find the parameters

```
## Your task: Make a fit to the Michealies-Menten equation for the data without the
inhibitor.
initial_guess = [Vmax_no_inhib_estimate, K_M_no_inhib_estimate]
fitted_parameters, trash = ...
vmax_fit_no_inhib, km_fit_no_inhib = fitted_parameters

## Your task: Make a fit to the Michealies-Menten equation for the data the inhibitor.
initial_guess = [Vmax_inhib_estimate, K_M_inhib_estimate]
fitted_parameters, trash = ...
vmax_fit_inhib, km_fit_inhib = fitted_parameters

## Printing
print('Without inhibitor:')
print('    V_max:', vmax_fit_no_inhib)
print('    K_M:', km_fit_no_inhib)
print('With inhibitor:')
print('    V_max:', vmax_fit_inhib)
print('    K_M:', km_fit_inhib)
```

Which of the kinetic parameters changes substantively in the presence of the inhibitor

(d) Analyze the fit

As per usual, we need to plot the fit to confirm that that it has been done successfully.

We start by evaluating the fit for each set of parameters - those with and those without the inhibitor.

```
## Your task: Evaluate the fits for both cases
S_smooth = np.linspace(0.0, 0.055)
```

```
V0_no_inhib_model = ...
V0_inhib_model = ...
```

Copy your code from above and then we plot the fit along with the data,

```
## Your task: Copy your plotting code from above and add the two lines to plot the fits.
```

(e) Inhibitor type

What type of inhibitor is inhibitor1?

(f) K_i

The experiment was done in the presence of 2.5 mM of the inhibitor. What is the K_i for this inhibitor?

```
## Your task: Set the inhibitor concentration in M.
I_concentration = ...

## Your task: Calculate K_i from the fitted K_M values
K_i = ...

print(K_i)
```

---

2 Enzyme inhibitors (II)

```
import matplotlib.pyplot as plt
import pandas as pd
from scipy.optimize import curve_fit
import numpy as np
from IPython.display import display
pd.set_option('display.max_rows', 6)
```

Two enzyme inhibitors were identified as part of a drug discovery program. To characterize the mechanism of action, the reaction kinetics were analysed for the enzyme alone and in the presence of 5 μM of each of the inhibitors as a function of substrate concentration resulting in the data in the file `enzyme-inhib-ii.xlsx`

Load the dataset

```
data_file_name = ... # Write the  of the uploaded data file
df = pd.read_excel(data_file_name)
display(df)
```

(a) Convert & plot

Convert the measurements to SI units and plot the initial reaction velocity versus substrate concentration for all three reactions.

Start by converting the units

```
## Your task: Add new columns with the properties in the correct SI units.
## There are 4 properties, so you should add 4 columns.
... # Your code to convert units here.
```

And then plot to visualize the data

```
fig, ax = plt.subplots()

## Your task plot each data vs. [S]_(M).
## To distinguish the data its a good idea to use label='enz_(M/s)' and so on.
...
...
...

ax.legend()
ax.set_xlabel('Substrate concentration [M]', fontsize=14)
ax.set_ylabel('$V_0$', fontsize=14)
plt.show()
```

(b) Estimate parameters

Based on the appearance of these plots:

- Which dataset would have the highest $V_{\max}$, which the lowest?
- What about K_M?

Based on your answers,

- Can you determine the type of inhibition?
- Alternatively, can you exclude some mechanisms?

(c) Fit

As always, we need the function we're fitting with

```
def michaelis_menten(S, Vmax, Km):
    return Vmax * S / (S + Km)
```

And then we can make the fit

```
## Your task fit dataset 1: 'enz_(M/s)'
initial_guess = [..., ...]
fitted_parameters, trash = curve_fit(michaelis_menten, df['[S]_(M)'], df['enz_(M/s)'],
initial_guess)
Vmax_enz_fit, K_M_enz_fit = fitted_parameters

## Your task fit dataset 2: 'inhibitor2_(M/s)'
initial_guess = [..., ...]
fitted_parameters, trash = curve_fit(michaelis_menten, df['[S]_(M)'], df['inhibitor2_(M/
s)'], initial_guess)
Vmax_inhib2_fit, K_M_inhib2_fit = fitted_parameters

## Your task fit dataset 3: 'inhibitor2_(M/s)'
initial_guess = [..., ...]
fitted_parameters, trash = curve_fit(michaelis_menten, df['[S]_(M)'], df['inhibitor3_(M/
s)'], initial_guess)
Vmax_inhib3_fit, K_M_inhib3_fit = fitted_parameters

print('Dataset 1: enz_(M/s)')
print('    V_max', Vmax_enz_fit)
print('    K_M', K_M_enz_fit)
print('Dataset 2: inhibitor2_(M/s)')
print('    V_max', Vmax_inhib2_fit)
print('    K_M', K_M_inhib2_fit)
print('Dataset 3: inhibitor3_(M/s)')
print('    V_max', Vmax_inhib3_fit)
print('    K_M', K_M_inhib3_fit)
```

```
fig, ax = plt.subplots()

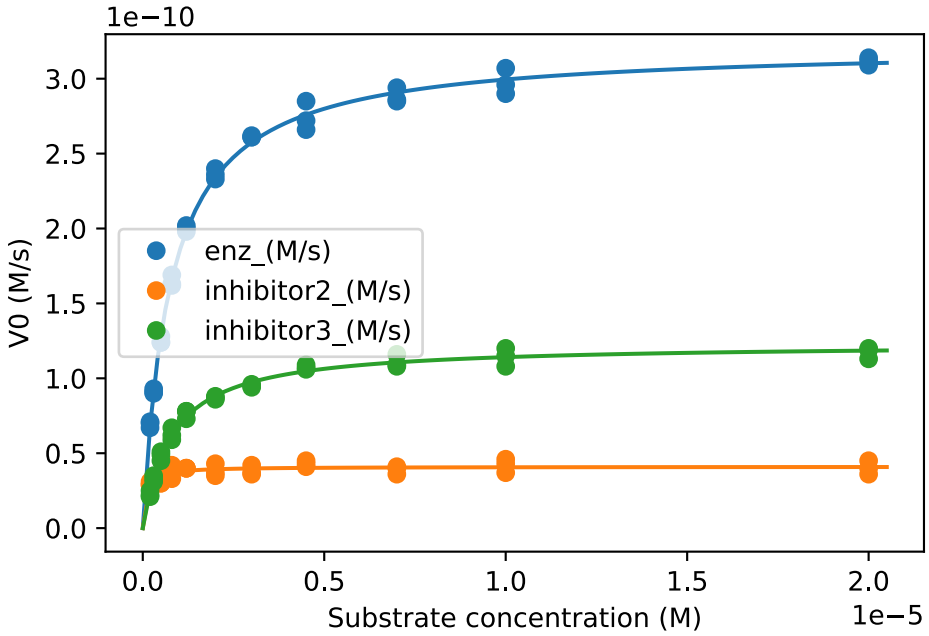
## Your task: Evaluate the fits
S_smooth = np.linspace(0, 2.05*10**(-5), 100)
V0_enz = michaelis_menten(S_smooth, Vmax_enz_fit, K_M_enz_fit)
V0_inhib2 = michaelis_menten(S_smooth, Vmax_inhib2_fit, K_M_inhib2_fit)
V0_inhib3 = michaelis_menten(S_smooth, Vmax_inhib3_fit, K_M_inhib3_fit)

## Plots the fits
```

```
ax.plot(S_smooth, V0_enz)
ax.plot(S_smooth, V0_inhib2)
ax.plot(S_smooth, V0_inhib3)

## Plots the data
ax.plot(df['[S]_(M)'], df['enz_(M/s)'], 'o', label='enz_(M/s)', color='C0')
ax.plot(df['[S]_(M)'], df['inhibitor2_(M/s)'], 'o', label='inhibitor2_(M/s)', color='C1')
ax.plot(df['[S]_(M)'], df['inhibitor3_(M/s)'], 'o', label='inhibitor3_(M/s)', color='C2')

## Adds legend and axis labels.
ax.legend()
ax.set_xlabel('Substrate concentration (M)')
ax.set_ylabel('V0 (M/s)')
plt.show()
```



If your fits don't closely match the data, go back and change your initial guesses.

(d) Type of inhibitor

We call $V_{\max}$ and K_{M} recorded in the presence of the inhibitor for $V_{\max}^{app}$ and K_{M}^{app} (app = apparent).

Calcuate how much each value changes in the presence of the inhibitor by dividing them by the parameters determined in the absence of an inhibitor.

```
## Your task: Calculate the ratio nof K_M^app / K_M
K_M_ratio2 = K_M_inhib2_fit / K_M_enz_fit
K_M_ratio3 = K_M_inhib3_fit / K_M_enz_fit

## Your task: Calculate the ratio nof Vmax^app / Vmax
Vmax_ratio2 = Vmax_inhib2_fit / Vmax_enz_fit
```

```
Vmax_ratio3 = Vmax_inhib3_fit / Vmax_enz_fit

## Prints the ratios
print('Inhibitor 2')
print('K_M', K_M_ratio2)
print('Vmax', Vmax_ratio2)

print('Inhibitor 3')
print('K_M', K_M_ratio3)
print('Vmax', Vmax_ratio3)
```

Based on the above, what is the likely inhibition type of inhibitor2 and inhibitor3?

(e) K_i

Determine the K_i for each of the two inhibitors. The inhibitor concentration is 5 μM.

```
## Your task: Assign known value
I = ...

## Your task: Calculate K_I for both cases.
## You should derive the correct expression before doing any coding!
K_I_inhib2 = ...
K_I_inhib3 = ...

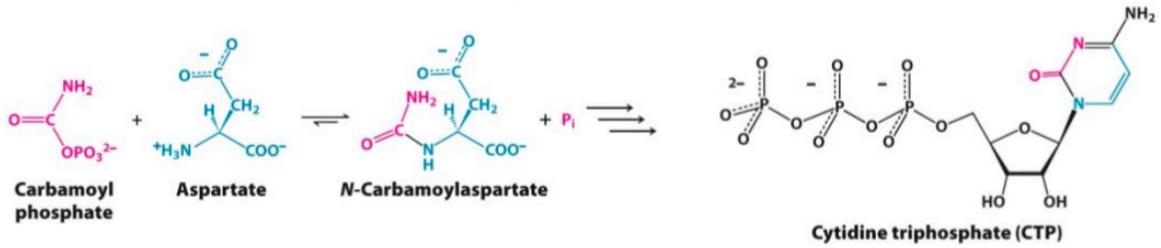
## Prints the results
print('Inhibitor 2: ', K_I_inhib2)
print('Inhibitor 3: ', K_I_inhib3)
```

---

3 Enzymatic behaviour of the enzyme ATCase

```
import matplotlib.pyplot as plt
from scipy.optimize import curve_fit
import numpy as np
import pandas as pd
from IPython.display import display
```

The enzyme aspartate transcarbamoylase (ATCase) catalyzes the first reaction in the biosynthesis of pyrimidines such as CTP as shown in the reaction below:



ATCase does not obey the Michaelis-Menten kinetics model but instead shows the behaviour recorded in the `enzyme-behav-atcase.csv` dataset.

The dataset consists of three columns; the aspartate concentration in mM and the rate of formation of N-carbamyolaspartate with and without the presence of CTP.

Load the dataset with pandas.

```
data_file_name = ... # Write the  of the uploaded data file
df = pd.read_csv(data_file_name)
display(df)
```

(a) Units & Plot

```
## Your task: Add columns with the properties in SI units.
df['[aspartate]_(M)'] = ...
df['rate_(M/s)'] = ...
df['rate_ctp_(M/s)'] = ...
```

Make a plot of the `rate_(M/s)` dataset versus the aspartate concentration.

```
fig, ax = plt.subplots()

## Your task: Plot the dataset
...

## Adds legend and labels.
ax.set_xlabel('[Aspartate] (M)')
ax.set_ylabel('Rate of formation (M/s)')
ax.legend()
```

(b) Kinetic profile

Looking at the curve without CTP, describe the kinetic profile of ATCase and explain what it tells us about the way ATCase works. (You may find inspiration in the material previously covered on protein-ligand interactions)

(c) Quarternary structure

What does the figure tell us about the quaternary structure of ATCase?

(d) Effect of CTP

Copy your plotting code from above and add a line to also plot the `rate_ctp_(M/s)` dataset.

```
## Your task: Copy your plotting code and add a line to also plot the rate_ctp_(M/s)
dataset.
...
```

Qualitatively describe the effect of CTP on the rate of N-carbamyolaspartate formation

(e) Physiological advantage

In fact many enzymes are regulated by certain end products in a fashion similar to the CTP effect on ATCase. Can you explain why this might be a physiological advantage?

(f) Regression with Hill equation.

There is also a formulation of the Hill equation for the enzyme kinetics, which has the following form:

$$v = \frac{V_{\max} \cdot S^h}{K_{\frac{1}{2}}^h + S^h}$$

Note, that we use a formulation of the Hill equation, where the constant on the denominator is raised to the power of h. Therefore, the constant also marks the half-way saturation concentration and we denote it $K_{\frac{1}{2}}$

As usual start by writing a function implementing this equation

```
def hill_eq(S, Vmax, K, h):
    ## Your task: Implement the enzyme kinetics Hill equation.
    result = ...
    return result

print(hill_eq(1, 1, 1, 1)) # Should give 0.5 -> (1 * 1^1) / (1**1 + 1**1) = (1 * 1) / (1 +
1) = 1/2
```

And then make the fit for each dataset of rates vs. aspartate concentration.

```
## Your task: Make the fit for the dataset without CTP
fitted_parameters_noctp, trash = ...
Vmax_fit_noctp, K_fit_noctp, h_fit_noctp = fitted_parameters_noctp

## Your task: Make the fit for the dataset with CTP
fitted_parameters_ctp, trash = ...
Vmax_fit_ctp, K_fit_ctp, h_fit_ctp = fitted_parameters_ctp
```

The next cell prints the fitted parameters for both cases.

```
print('Without CTP')
print('    Vmax', Vmax_fit_noctp)
print('    K1/2', K_fit_noctp)
print('    h', h_fit_noctp)

print('With CTP')
print('    Vmax', Vmax_fit_ctp)
print('    K1/2', K_fit_ctp)
print('    h', h_fit_ctp)
```

Then we can plot

```
fig, ax = plt.subplots()

## Your task: Calculate the fits
S_smooth = np.linspace(0, 50*10**(-3))
V_fit_noctp = ...
V_fit_ctp = ...

## Your task: Plot the fits:
...
...

## Plots the datasets
ax.plot(df['[aspartate]_(M)'], df['rate_(M/s)'], 'o', label='No CTP', color='C0')
ax.plot(df['[aspartate]_(M)'], df['rate_ctp_(M/s)'], 'o', label='CTP', color='C1')

## Labels & legend
ax.set_xlabel('[Aspartate] (M)')
ax.set_ylabel('v (M/s)')
ax.legend()
```

How do the constants of the Hill equation change in response to the presence of CTP?
Does this match your expectations?